The BB(6) Program

Machines, machine-checked proofs, and incision geometry on the Busy Beaver frontier

`1RB1RA_0LC1LE_1LD1LC_1LA0LB_1LF1RE_---0RA`

BB6

bbchallenge-proofs · 2026-08-22

The game: Busy Beaver

Among all n-state Turing machines started on a blank tape, which runs longest before halting? BB(n) outgrows every computable function — it is the edge of what computation can know.

- BB(5) settled July 2024, machine-verified in Rocq (Coq)
- BB(6) is the frontier: value exceeds numbers writable in this universe
- “Cryptids”: machines whose halting encodes deep open arithmetic

47,176,870

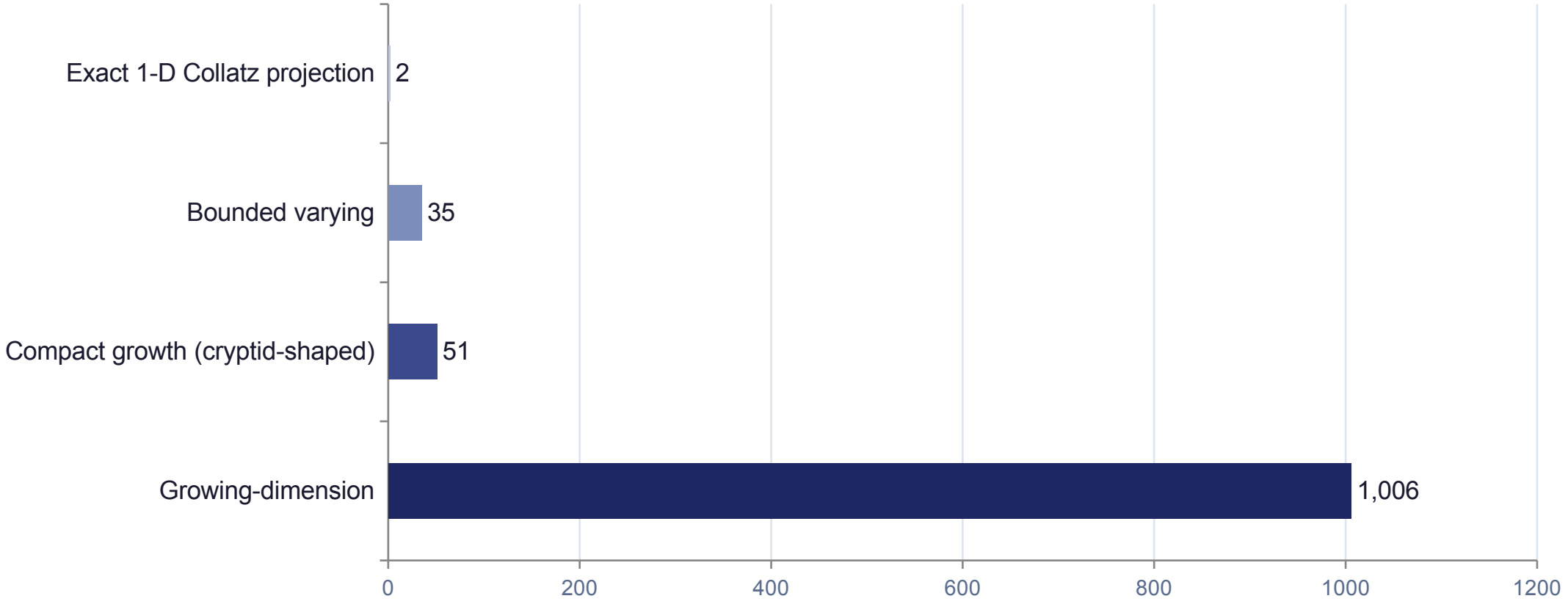
steps — BB(5), proven exactly

1,158

BB(6) machines still undecided

What the open set looks like

Every holdout classified through the Collatz-lens record structure (stage745, 0 errors). Collapse-like behavior is the dominant phenotype — and exactly two machines are literal 1-D Collatz.



Exact emulation, fail-closed

A macro system is a set of arithmetic rules: “from a record of shape X, exactly f(k) raw steps reach shape Y.” Rules must replay the raw machine perfectly — or the fit is rejected.

100%

exact replay, in & out of sample, over 10^8 raw steps

13 / 13

Antihydra rule obligations discharged (stage753)

33 / 33

dossier-1 rule obligations discharged

6 + 1

solid + weak exact-macro passes after the 2026-08-04 gate fix

Two machines, formally reduced

A

Antihydra

`1RB1RA_0LC1LE_1LD1LC_1LA0LB_1LF1RE_---0RA`

```
safeOrbitA_nonhalting :  SafeOrbitA → ¬ halts
```

Three record rules (hydra / drop / merge) proven for all block values; 62-step startup; orbit conjecture = the community's Antihydra conjecture.

D

Dossier-1

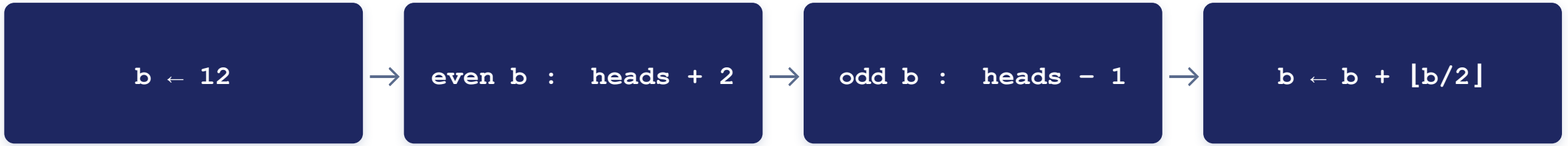
`1RB0LE_1LB1RC_0RF0RD_1RE0LA_1LA1RE_---0RB`

```
safeOrbit_nonhalting :  SafeOrbit → ¬ halts
```

Seven rules over a stack of chain levels with collapse cascades; 35-step startup; conjecture is a recursive (3n+7)/2 checkpoint law.

The bridge to the real conjecture

The community states Antihydra's conjecture 1-dimensionally. A machine-checked conjugacy proves our orbit hypothesis is exactly that statement (one boundary case stronger).



The change of variables

```
b  =  m + 2·(c−2)/3 + 5
heads  =  a   (the machine's counter block)
merges are the even steps, drops the odd ones
```

The honest delta

Formal death fires at heads ≤ 1; the wiki's at 0. The tape grammar needs a non-empty counter block, so the hypothesis is one drop stronger — documented, and closing it is listed follow-up work.

Check only where the cut can happen

- Death set is explicit: m = 0, or a drop (m = 2) with counter ≤ 1
- Volley: every hydra run collapses to one closed-form jump
- Event process = one boundary rule + one volley, per counter move
- Theorem: event process dies ⇔ full orbit dies (sound AND complete)

16

incision events cover the first 10^8 raw machine steps

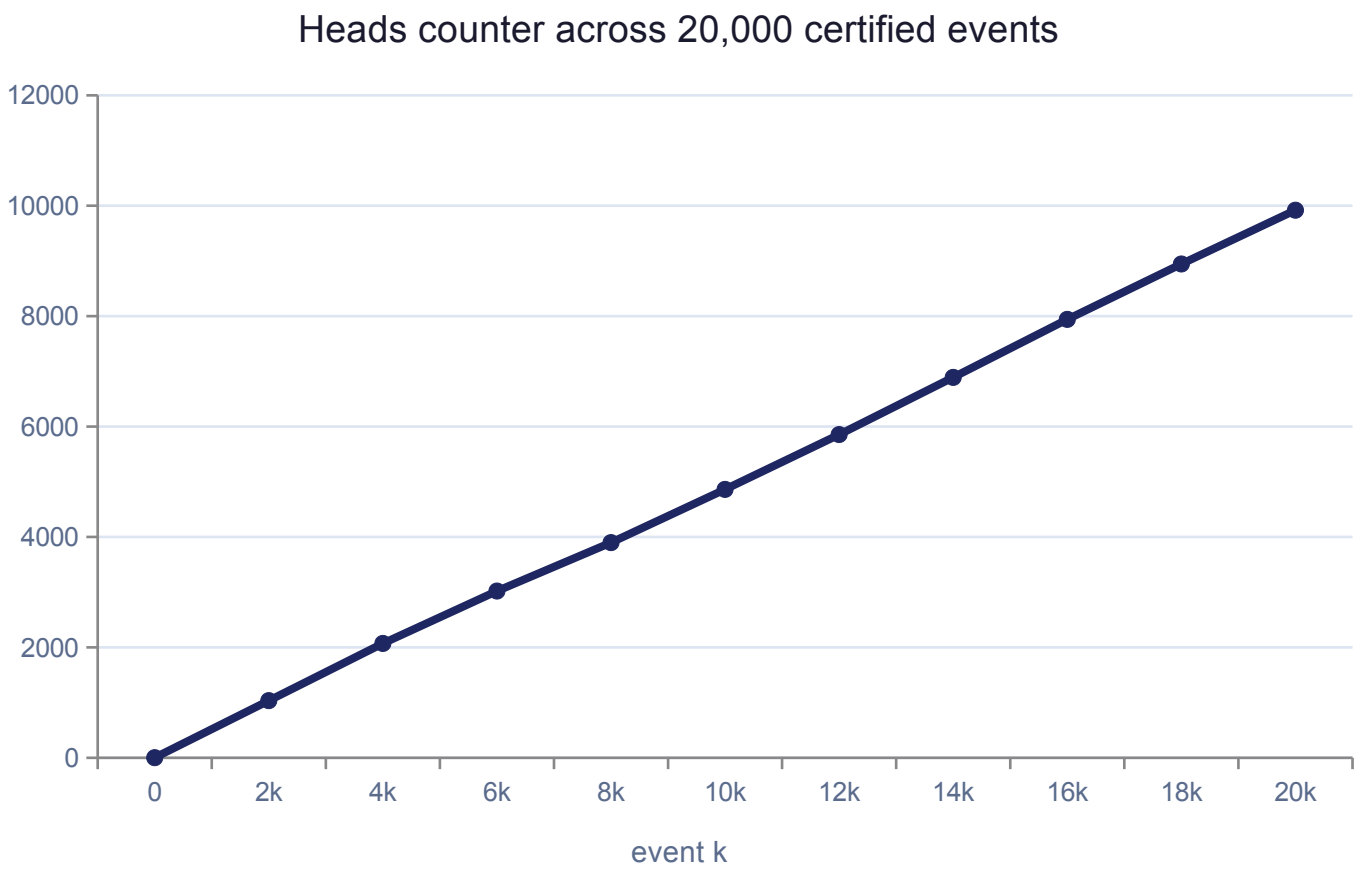
`eventSafe_iff_safeOrbitA`

certified in core Lean · axioms [propext, Quot.sound]
chained: event safety alone ⇒ Antihydra never halts

10^{7046}

raw-step equivalent of 20,000 events — a different regime, not an optimization

The incision atlas (stage756)



+0.5 / event

measured counter drift — exactly the random-walk expectation

margin 4

closest approach to the death condition in 20k events

3 + 2

machines pairing up into identical incision profiles (delta-rule families)

Trust engineering

1. **Axiom gate**

   Every theorem reports exactly [propext, Quot.sound]; a script fails the build otherwise. Two contamination channels found and engineered out (omega on compound goals; String.toList).

2. **Parser parity**

   A Lean parser of the standard machine format is kernel-proved equal to the hand-written tables — with corrupted-code negative controls.

3. **Differential testing**

   The proved step function, run executably, is byte-identical for thousands of steps against an independent 30-line Python stepper.

4. **Fail-closed corrections**

   Unsound certificates demoted; the PASS gate retightened and two verdicts publicly refuted. Corrections are results, not embarrassments.

Validated for Tao's Palomar registry

Palomar (announced 2026-08-18): a registry of Lean-verified mathematics. Small auditable Challenge + proving Solution, mechanically compared by kernel replay, three permitted axioms.

Antihydra

120-line challenge · machine + hydra process

`antihydra_nonhalting`

Incision

79-line challenge · pure arithmetic

`boundary_checking_complete`

Dossier-1

153-line challenge · machine + chain orbit

`dossier1_nonhalting`

Every mechanical intake check passes — solutions transport the certified theorems across self-contained challenge definitions. Remaining: a LICENSE choice, a git push, and owner sign-off.

Where this stands

PROVEN

Every dynamical fact about both machines · the reductions · the bridge · the incision theorem · parser parity

ASSUMED

Two orbit conjectures — one is the Antihydra conjecture itself. Believed true; proven by no one.

NEXT

Close the heads ≤ 1 boundary case · dossier-1's 1-D bridge · log-scaled atlas · Palomar submission on sign-off

Nothing here decides any machine — each machine's fate is now a clean piece of arithmetic, plus the certified tools to hunt it.